



Title of the Project:

Blood Bank Management System

Submitted by,

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ACKNOWLEDGEMENT

Completing this project would not have been possible without the participation and assistance of many individuals. We would like to express our deep appreciation and indebtedness to our teachers and supervisors for their endless support, kindness, and understanding during the project duration.

Firstly, we are grateful to our principal, **Dr. Janardan Pawar**, for his support. We are also thankful to **Mr. Ninad Thorat** for their valuable guidance and support. Without their assistance, this project would not have been completed. This project has been a source of learning and has enabled us to apply our theoretical knowledge to the real world. Therefore, we acknowledge their help and guidance for this project. Once again, thanks to everyone for making this project successful.

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1) INTRODUCTION

1.1) Existing System and Need for System

In Existing System,

All Data of how many blood bags or quantity are available, to update those quantity and also the registering a donor by saving its details, updating it as required, searching for a donor in database and also deleting a donor from database. All these data is been stored and updated in the Database by writing queries .

There is not any user interface to manage the data of donor and blood stock.

It is very hectic to write each and one query to fetch the data, insert the data,

Update the data, delete the data and other Crud operations.

1.2) Scope of Work

For this project, I am going to create a desktop application which can handle the donor's data as well as blood stock.

I am going to create Frontend, Backend as well as database for my application.

Frontend :- For Frontend, I am going to use windows form Application in which C# Programming Language is used to create Frontend part. Generally, C# Windows Forms is a graphical user interface (GUI) framework that enables developers to create desktop applications for the Windows operating.

Database :- For database, I am going to MSSQL (Microsoft Structured Query Language) which we will use to store database.

Backend :- For Backend, I am going to use a Programming Language called C#. Which I will use to perform some crud operations and writing the logic integrating it with Database.

1.3) Operating Environment – Hardware and Software

Hardware Requirements:-

Processor – Core i3 or higher

Speed - 2 GHz & above.

RAM – 8.00 GB.

Hard Disk – 500 GB and above.

Software Requirements:-

Operating System: - Windows 10/11.

Windows Form Application.

C# Language.

2) PROPOSED SYSTEM

2.1) Proposed System

In the proposed system, we have following modules.

User Authentication Module :-

Manages login functionality for admins to access the system securely.

Donor Management Module :-

Allows the admin to register a new donor by entering their details.

Enables the admin to update donor details, such as name, contact information, and blood group.

Provides functionality to display donor details, allowing the admin to view all donors' information.

Implements the ability to delete a donor from the system.

Search Module :-

Enables the admin to search for donors by their blood group and location.

Retrieves donor information based on search criteria and displays the results.

Blood Stock Management Module :-

Manages the quantity of blood stock available.

Allows the admin to increase or decrease the quantity of blood stock as needed.

These modules will work together to provide the necessary functionalities for the admin to manage donors and blood stock effectively in your desktop application.

2.2) Objectives of System :-

The Objective of this Blood Bank Management System is that :-

- The user can add a donor simply by using this software.
- The user can update his/her details easily.
- This Software allows report generation and allows user to print it.
- The user can delete the donor if necessary with the help of user interface.
- The user can search the donor.
- The user can update the blood stock quantity
As needed.

2.3) User Requirements

The Blood Bank Management System is required for maintaining the data of the Blood Donors and stock

Requirements:-

1) Storing Data :-

User can Store Data of Blood Stock and Donor where he/she can register the user without writing the query directly into database.

2) Update Data:-

User can change the name, phone number, and all its details as needed by using this software.

3) Display the Data:-

User can Display Data of Blood Donor as well as Stock Quantity.

4) Search The Donor:-

User can search the Donor By his/her location and blood group.

5) Print the Document:-

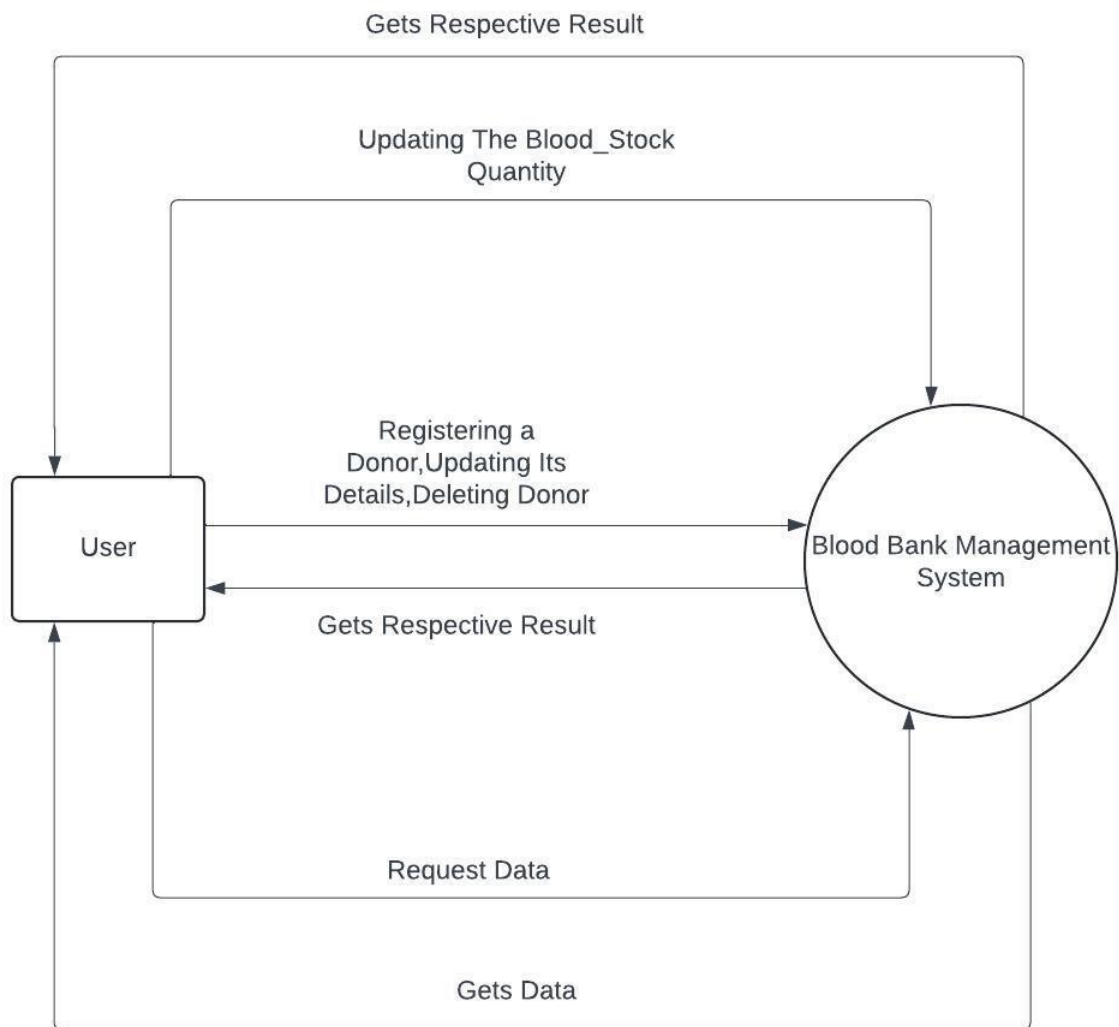
The Data Displayed on Screen should be able to get printed.

6) Delete the Donor:-

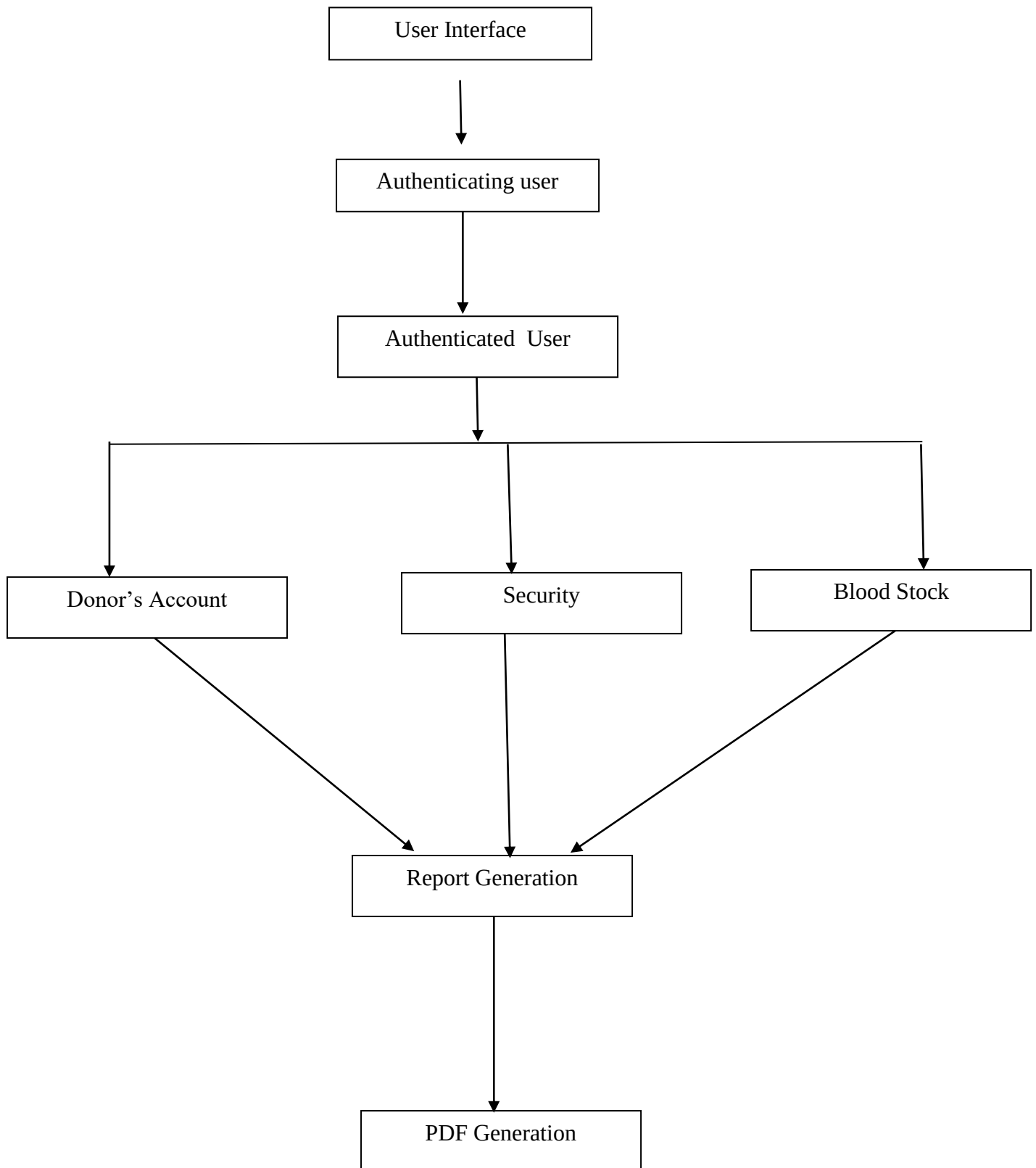
If needed, User can be able to delete the Donor by using his/her id.

3) ANALYSIS & DESIGN

3.1) Data Flow Diagram (DFD)

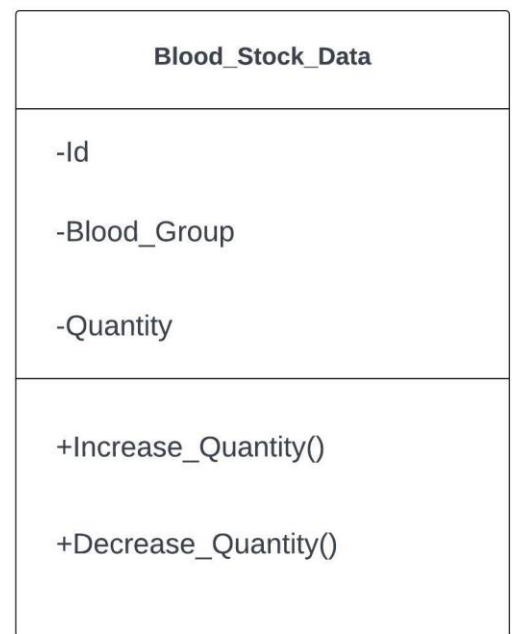
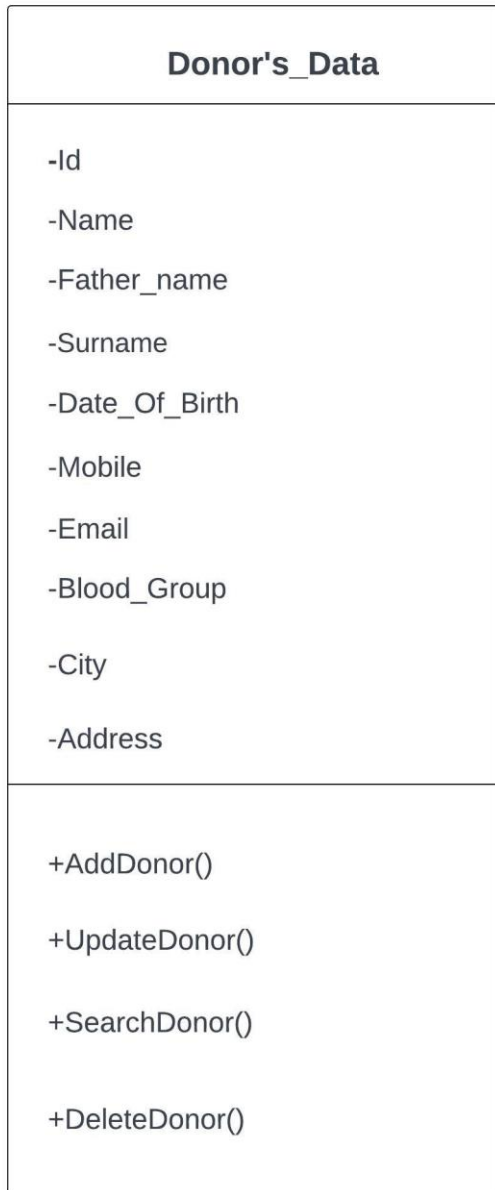


3.2) Functional Decomposition Diagram (FDD)



3.3) UML Diagrams:-

3.3.1) Class Diagram for Blood Bank Management system:-

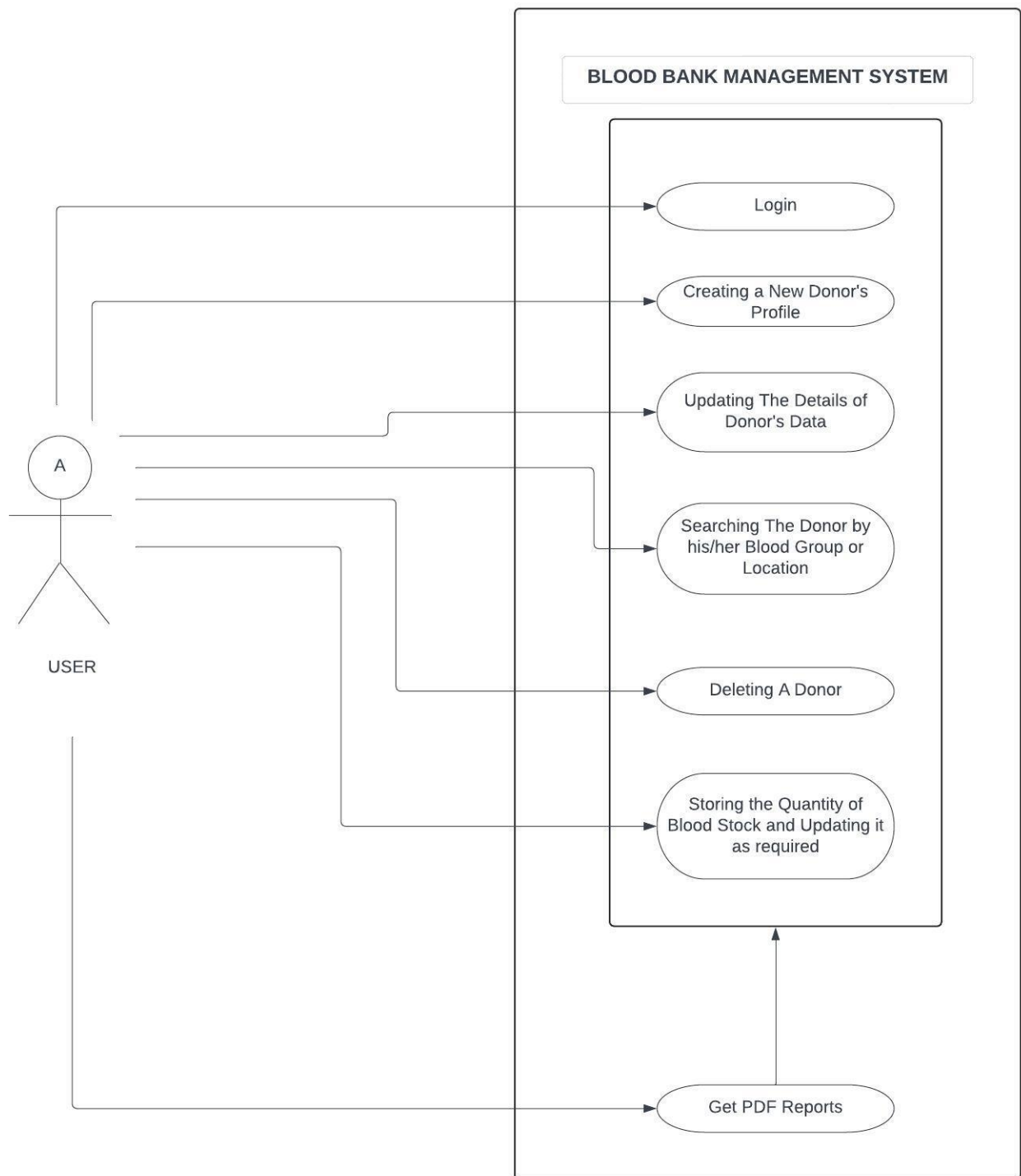


3.3.2) Object diagram for User Interaction:-

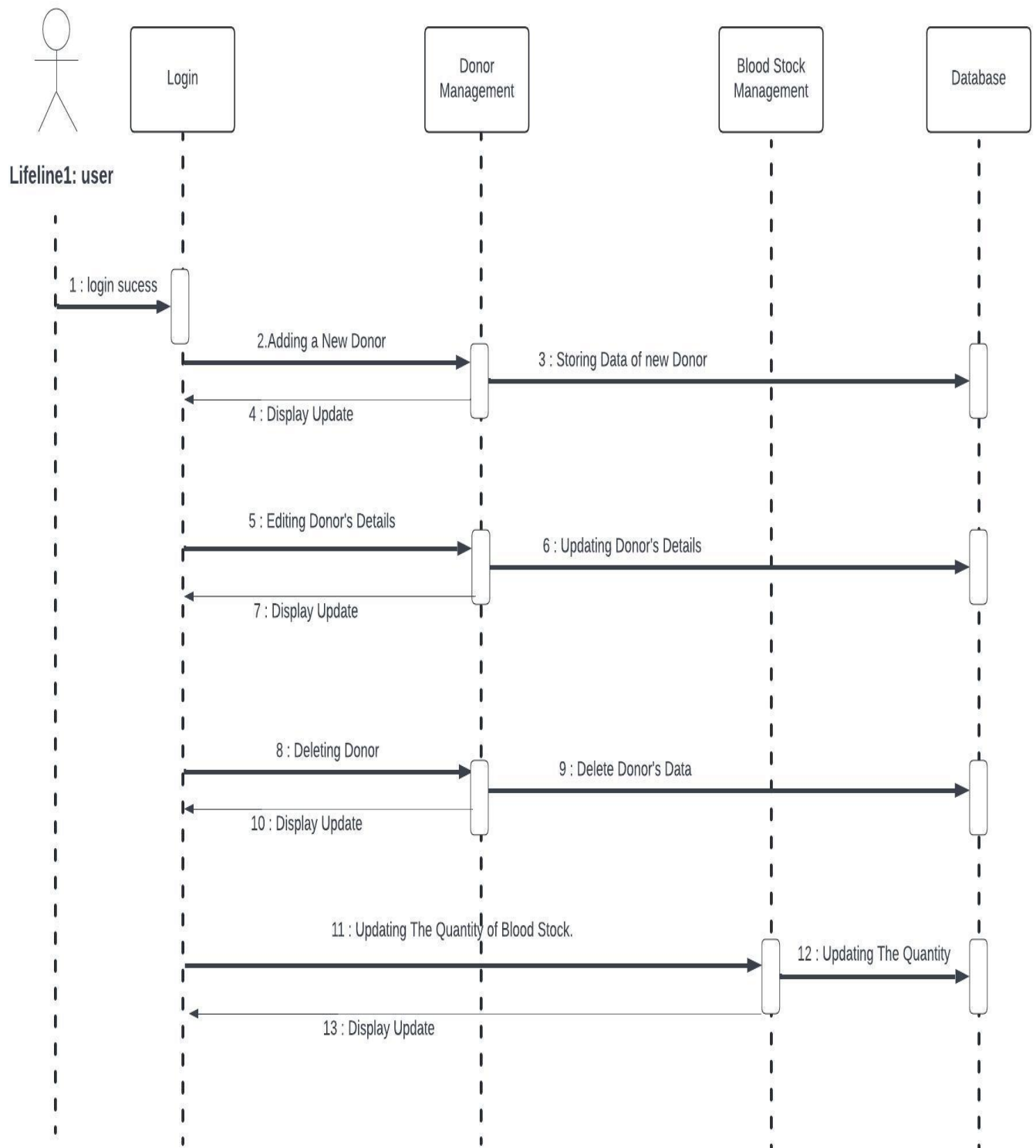
Donor's_Data
Id:4 name:Vikkram Father_name:Abhinav Surname:Saraswas Date_Of_Birth:1/15/1990 Mobile:9772744318 Email:vikram7890@gmail.com Blood_Group:O+ City:Pune Address:Flatno.404 Zinnea Appartments near Chandani Chowk Flyover
+AddDonor() +UpdateDonor() +SearchDonor() +DeleteDonor()

Blood_Stock_Data
Id:1 Blood_Group:A+ Quantity:130
+Increase_Quantity() +Decrease_Quantity()

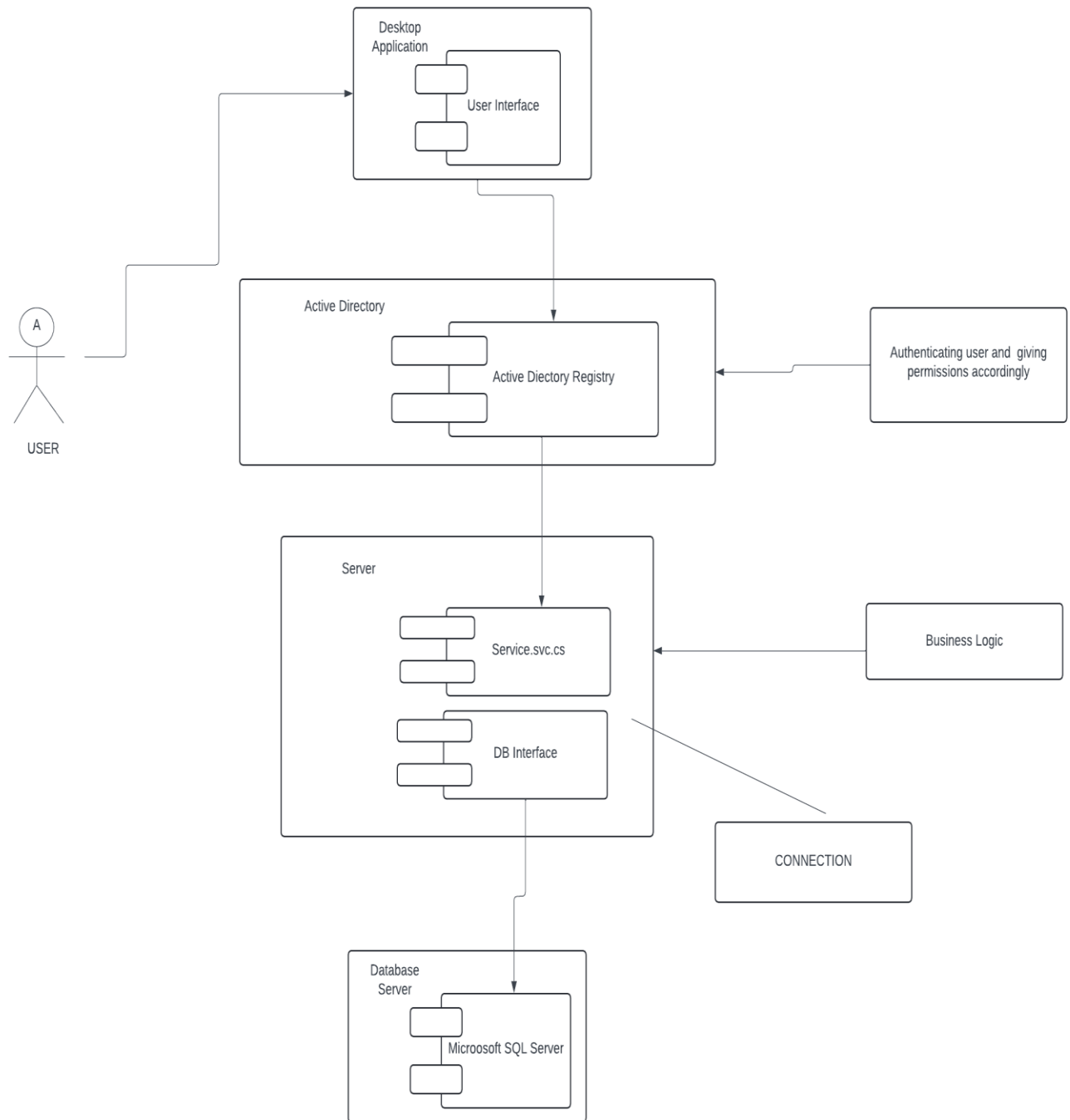
3.3.3) Use Case Diagram for Blood Bank Management system:-



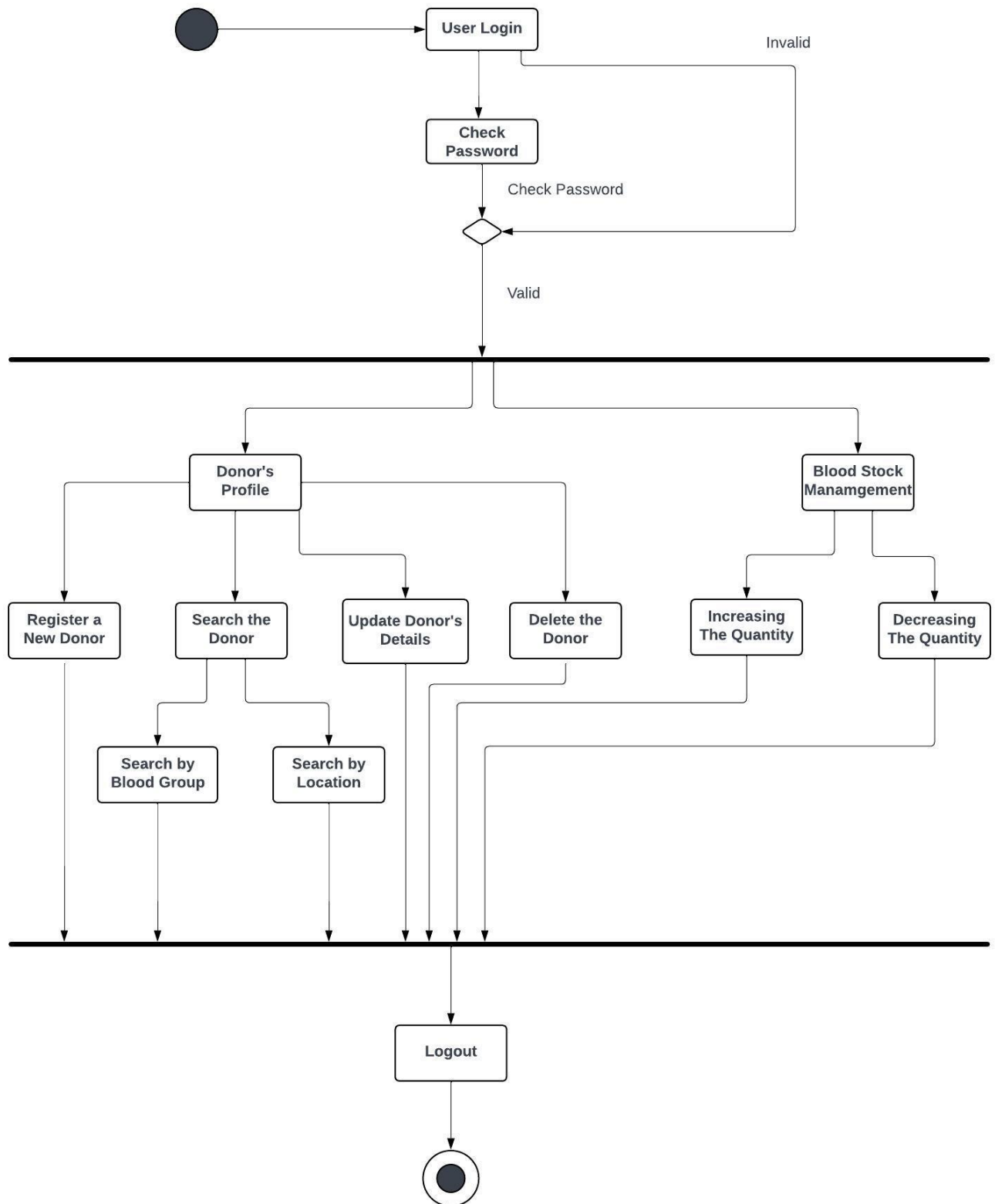
3.3.4) Sequence Diagram for Blood Bank Management system:-



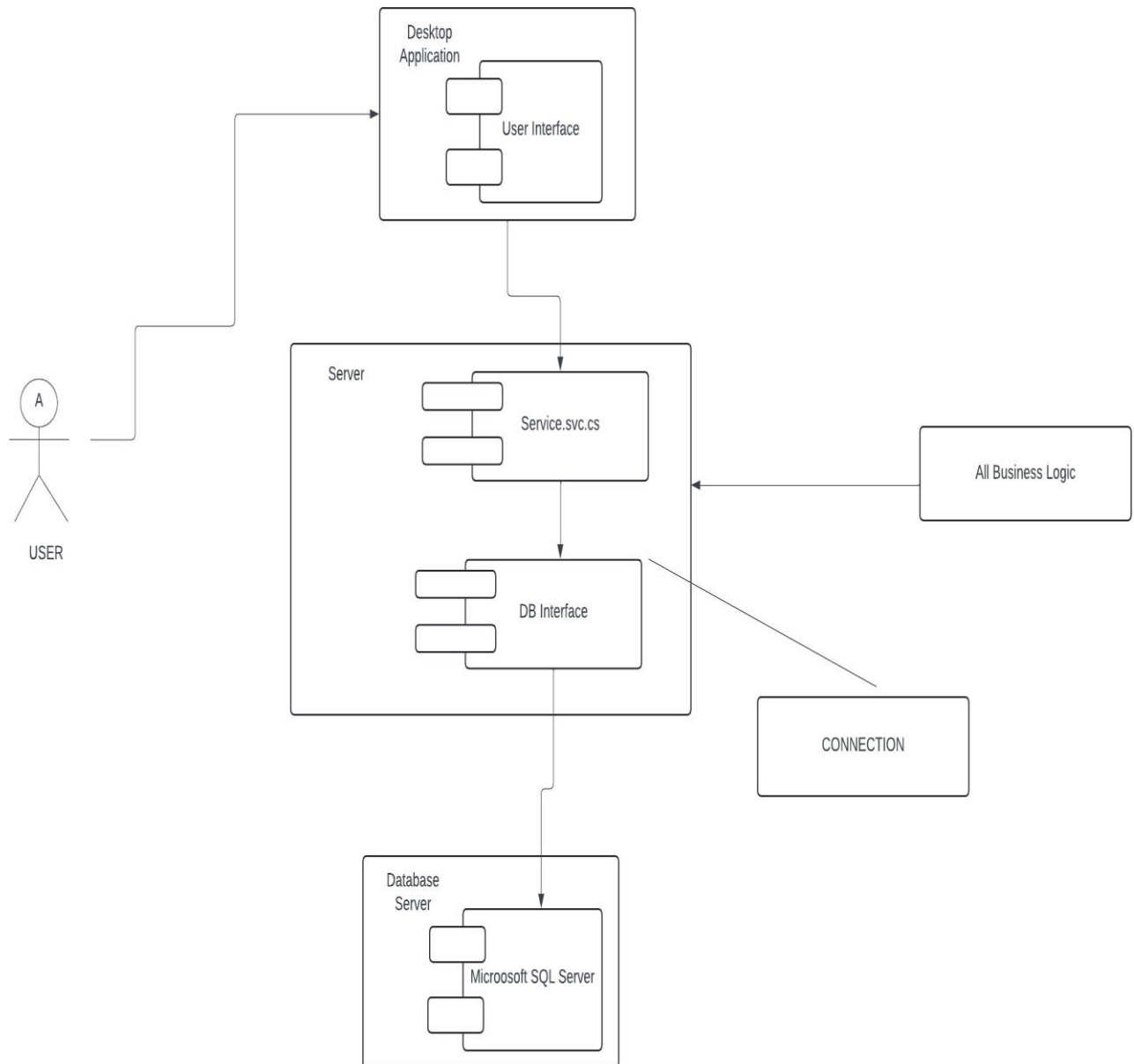
3.3.5) Collaboration_Diagram_for_Portfolio_Management system:-



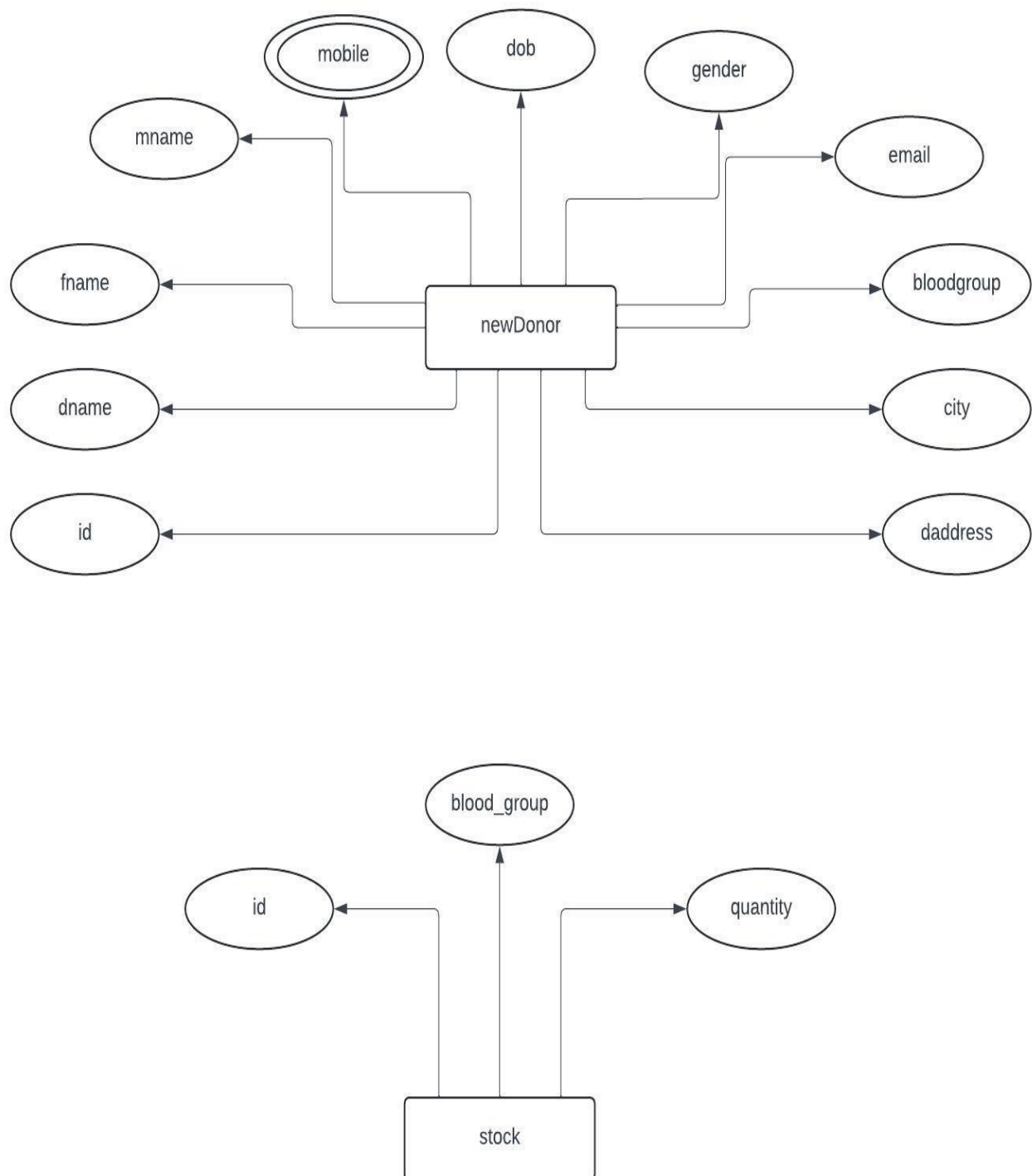
3.3.6)Activity Diagram for Port folio Management system:-



3.3.7)Deployment Diagram for Blood Bank Management system:-



3.4) Entity Relationship Diagram:-



3.5) Data Dictionary

Database :- bloodbank

Table Name :- newDonor

Column Name	Data Type	Constraints	Description
did	Int (4)	Primary Key	
dname	Varchar (250)		
fname	Varchar (250)		
mname	Varchar (250)		
dob	Varchar (50)		
mobile	Bigint (8)		
Gender	Varchar (25)		
email	Varchar (250)		
bloodgroup	Varchar (20)		
City	Varchar (150)		
daddress	Varchar (550)		

Database :- bloodbank

Table Name :- stock

Column Name	Data Type	Constraints	Description
bid	Int (4)	Primary Key	
blood_group	Varchar (50)		
quantity	Varchar (50)		

3.6) Table Design

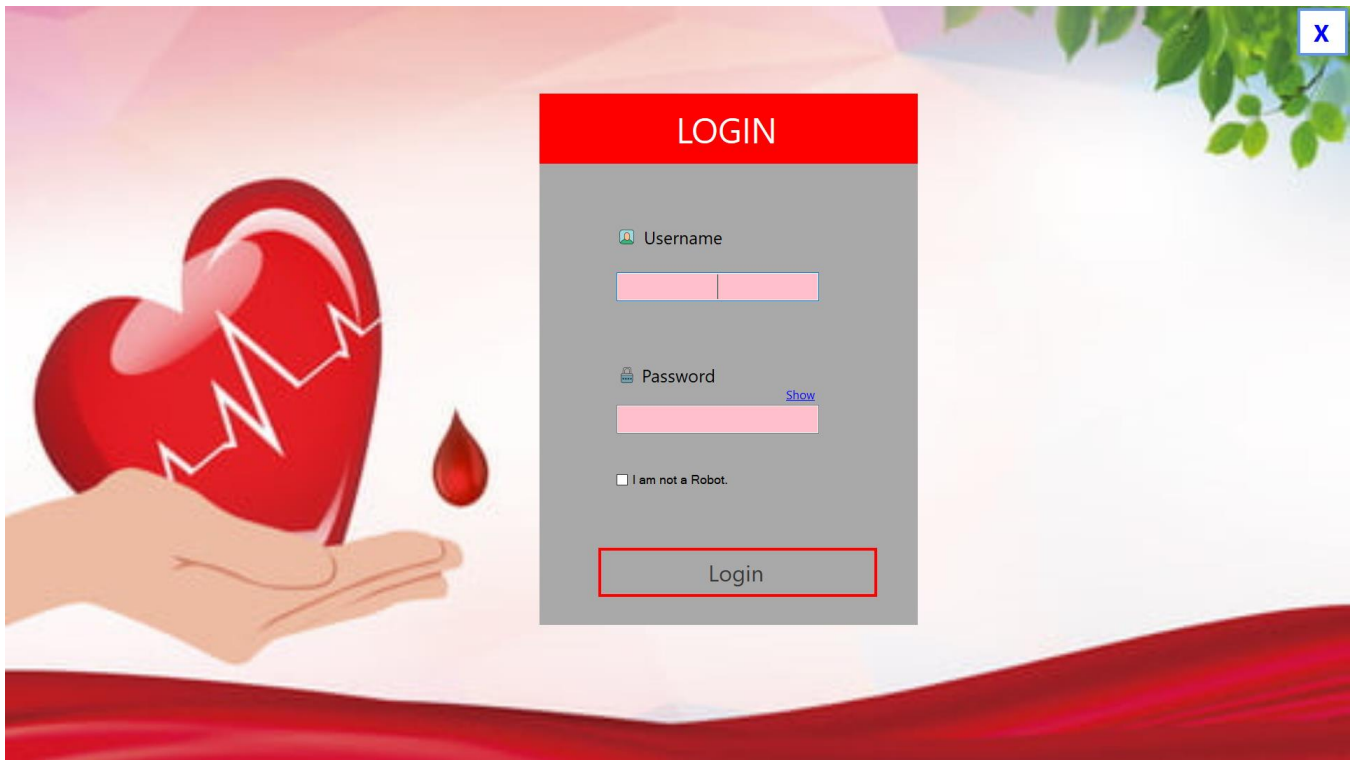
newDonor	
	did
	dname
	fname
	mname
	dob
	mobile
	gender
	email
	bloodgroup
	city
	daddress

stock	
	bid
	blood_group
	quantity

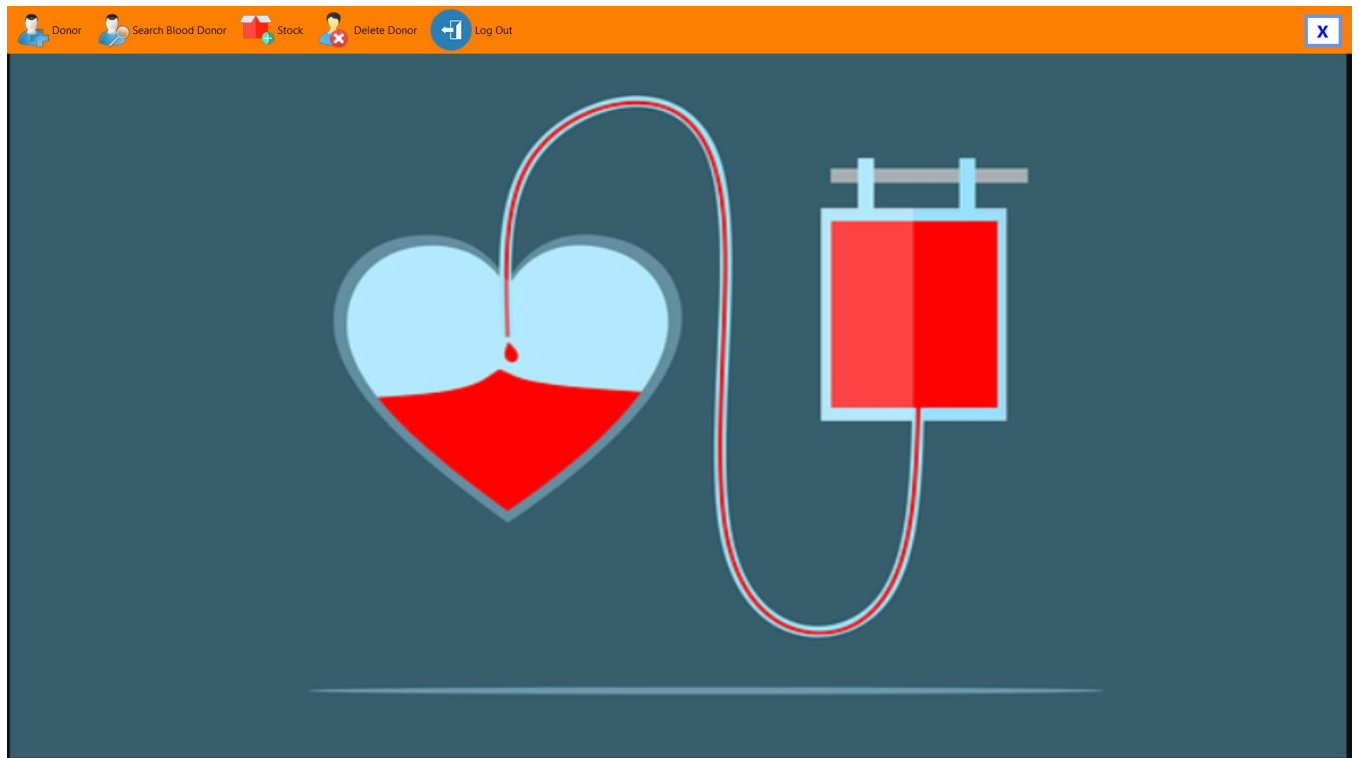
4)USER MANUAL

4.1) Screens

1)Login Screen :



2: Dashboard Screen :



3)Adding a New Donor :

The 'Add New Donor' form is a modal window with an orange background and a textured surface. It is titled 'ADD NEW DONOR' in bold black text. The form contains the following fields and controls:

- New Donor ID:** 20 (displayed in green text)
- Name:** Text input field
- Father Name:** Text input field
- Surname:** Text input field
- Date Of Birth:** Date picker showing 4/18/2024
- Mobile Number:** Text input field
- Gender:** Dropdown menu
- E-mail:** Text input field
- Blood Group:** Dropdown menu
- City:** Text input field
- Address:** Text area

At the bottom of the form, there are three buttons: 'SAVE' (with a floppy disk icon), 'RESET' (with a circular arrow icon), and 'CLOSE' (with a red X icon).

4) Donor's Details

ALL DONOR DETAILS

did	dname	fname	mname	dob	mobile	gender	email	bloodgroup	city	daddr
4	Vikram	Abhinav	Saraswas	1/15/1990	9772744318	Male	vikram7890@gmail.com	O+	Pune	Flatno.
5	Sagar	Ramesh	Pansare	10/25/2000	8944031120	Male	sagart567@gmail.com	A+	Pune	Bavdha
6	Priya	jogendare	Agarwal	3/7/2024	8900613351	Female	priya456@gmail.com	O+	Delhi	Flatno.
8	Vinita	Singh	Rajput	12/26/1995	9055014431	Male	Vinitaraj@gmail.com	A-	Jodhpur	House
9	Anjali	Vikram	Shinde	6/30/1969	9077461191	Female	anjali569a@gmail.com	O+	Nashik	vishap
10	Abhishek	Dilip	Nale	1/16/2009	8755939303	Male	abhishek@gmail.com	O+	Pune	pimpri
11	Akash	Arjun	chaudhari	3/14/2024	903786485290	Male	akashac@gmail.com	AB+	pune	Sesh b
13	usj	ks	nsik	3/19/2024	68902789	Male	ahoi@gmail.com	AB+	Pune	pune
14	897	ks	nsik	3/19/2024	68902789	Male	ahoi@gmail.com	AB+	Pune	pune
15	Danny	john	vickston	3/20/2024	8989	Male	asd	B+	pune	pune
16	fhf	sdfs	sdf	2/29/2024	89867	Female	fgod	O+	sddf	sdfs
19	Anthony	Vikram	Dsouza	6/5/1990	8799875591	Male	dsf@gmail.com	B+	pune	goa p

Print Close

5) Screen of Searching Donor By Location :

SEARCH DONOR BY LOCATION

Address

did	dname	fname	mname	dob	mobile	gender	email	bloodgroup	city	daddress
4	Vikram	Abhinav	Saraswas	1/15/1990	9772744318	Male	vikram7890@gmail.com	O+	Pune	Flatno.404
5	Sagar	Ramesh	Pansare	10/25/2000	8944031120	Male	sagart567@gmail.com	A+	Pune	Bavdhan B
6	Priya	jogendare	Agarwal	3/7/2024	8900613351	Female	priya456@gmail.com	O+	Delhi	Flatno.25,
8	Vinita	Singh	Rajput	12/26/1995	9055014431	Male	Vinitaraj@gmail.com	A-	Jodhpur	House no.
9	Anjali	Vikram	Shinde	6/30/1969	9077461191	Female	anjali569a@gmail.com	O+	Nashik	vishapur, n
10	Abhishek	Dilip	Nale	1/16/2009	8755939303	Male	abhishek@gmail.com	O+	Pune	pimpri chi
11	Akash	Arjun	chaudhari	3/14/2024	903786485290	Male	akashac@gmail.com	AB+	pune	Sesh build
13	usj	ks	nsik	3/19/2024	68902789	Male	ahoi@gmail.com	AB+	Pune	pune
14	897	ks	nsik	3/19/2024	68902789	Male	ahoi@gmail.com	AB+	Pune	pune
15	Danny	john	vickston	3/20/2024	8989	Male	asd	B+	pune	pune
16	fhf	sdfs	sdf	2/29/2024	89867	Female	fgod	O+	sddf	sdfs

Print Close

6) Screen of Searching Donor By Blood Group :

SEARCH DONOR BY BLOOD-GROUP

Blood Group:

did	dname	fname	mname	dob	mobile	gender	email	bloodgroup	city	daddress
4	Vikram	Abhinav	Saraswas	1/15/1990	9772744318	Male	vikram7890@gmail.com	O+	Pune	Flatno.404 Zinne
6	Priya	jogendare	Agarwal	3/7/2024	8900613351	Female	priya456@gmail.com	O+	Delhi	Flatno.25, Bajid
9	Anjali	Vikram	Shinde	6/30/1969	9077461191	Female	anjali569a@gmail.com	O+	Nashik	vishapur, naraya
10	Abhishek	Dilip	Nale	1/16/2009	8755939303	Male	abhishek@gmail.com	O+	Pune	pimpri chinchwa
16	fhf	sdfs	sdf	2/29/2024	89867	Female	fgod	O+	sdff	sdfs

Print Close

7)Blood Stock Details Screen:

STOCK DETAILS

bid	blood_group	quantity
1	A+	130
2	O+	219
3	B+	101
4	AB+	110
5	A-	97
6	O-	50
7	B-	70
8	AB-	90

* 1 Unit is Equal to 1 Blood Bag and 1 Blood Bag can contains upto 450 milliliters (mL) of Blood.

Print Close

8)Blood Stock Quantity Updation (Increasing/Decreasing)

Screen :

STOCK (INCREASE)

Blood Group Units

blood_group	quantity
A+	130
O+	219
B+	101
AB+	110
A-	97
O-	50
B-	70
AB-	90

Donor
Search Blood Donor
Stock
Delete Donor
Log Out

STOCK (DECREASE)

Blood Group
Units
Decrease

blood_group	quantity
A+	130
O+	219
B+	101
AB+	110
A-	97
O-	50
B-	70
AB-	90
*	

Print
Close

Deleting Donor :

Donor
Search Blood Donor
Stock
Delete Donor
Log Out

DELETE DONOR

Donor ID : 4
Search

Name
Vikram
E-mail
vikram7890@gmail.com

Father Name
Abhinav
Blood Group
O+

Surname
Saraswas
City
Pune

Date Of Birth
1/15/1990
Address
Flatno.404 Zinnes Apartments
near Chandani Chowk Flyover

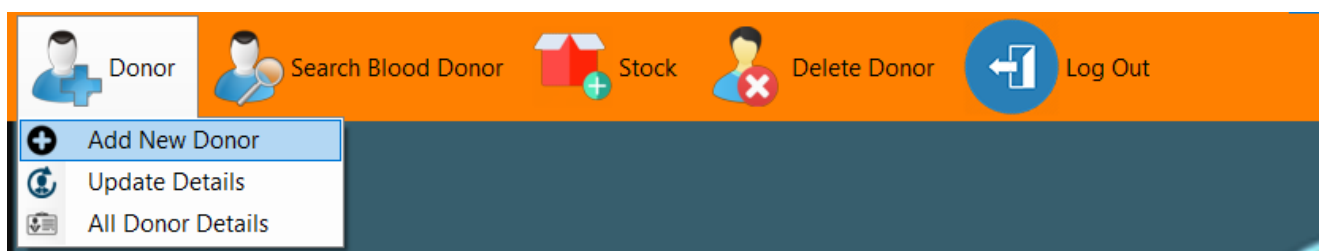
Mobile Number
9772744318

Gender
Male

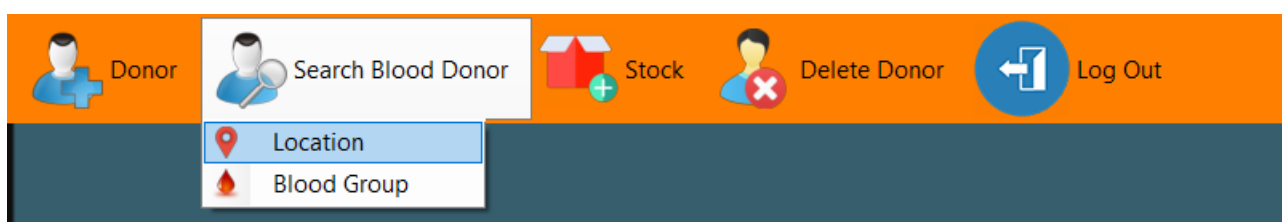
Delete
Reset
Close

4.2 Operations Manual / Menu Explanation:-

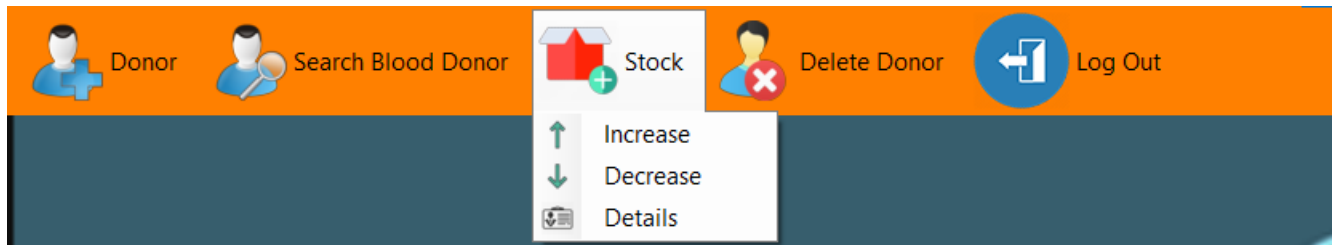
- You have to Login First.
- If you have login and entered into main (dashboard) screen.
- If you want to register a new donor you need to go to the donor section where you have to choose Add New Donor and then you can fill the details.



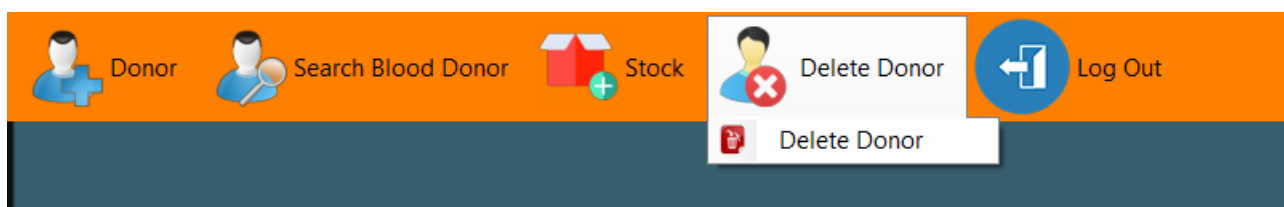
- If you want to update donor's details you need to go to the Update Details section as you can see in the above image.
- If you want to display the data you can go to the All Donor Details Section.



- If you want to search Donor by its Blood Group or Location you can go to search blood donor as shown in above Image.



- If you want to increase or decrease blood stock quantity you can go to Stock Section as shown in above image.



- If you want to delete donor you can go to delete donor section and write id and all the information will be displayed using search button and then you can click on delete button which will delete the donor.
- Then if your all operations are been done you can logout.

5) CONCLUSION

5.1) Drawbacks:-

- **Limited Cross-Platform Compatibility:**

Windows Forms Applications are primarily designed to run on the Windows operating system. They do not have native support for running on other platforms such as macOS or Linux. This limitation restricts the potential audience for applications developed using Windows Forms.

- **Outdated User Interface:**

Windows Forms uses a traditional UI framework that may appear outdated compared to more modern UI frameworks like WPF (Windows Presentation Foundation) or web-based technologies. As a result, Windows Forms applications may not provide the same level of aesthetic appeal or user experience as applications built with newer technologies.

5.2) Future Enhancement :-

- The user will be able to find that which Donor have donated how much Blood and at what time?
- The user will be able to track and keep records digitally where the Blood Bags (Stock) are going.
- The User will be able to keep the records of Health Related Details of each Donor.

CONCLUSION

This project gave me an opportunity to understand the basic flow of Blood Bank Management System as per the expectation of the client and the challenges one faces during the course of developing and implementing the project successfully. There was a time where a person need to write each and one query manually which takes more time but, now it become very easy to store the donor's data and update it time to time. Due to this user interface it becomes more easy to search a particular Donor and update stocks quantity as required. This Software makes the workflow much more efficient and easy.

BIBLIOGRAPHY

1) C# Language :-

- I. “C# 2010 IN SIMPLE STEPS” Authored By Kogent Learning SolutionsInc.
- II. <https://www.javatpoint.com/c-sharp-tutorial/>

2) SQL:

Website Link :- <https://www.w3schools.com/sql/>

Windows Form Application:

- I. Official Documentation of Microsoft:-
<https://learn.microsoft.com/en-us/visualstudio/ide/create-csharp-winform-visual-studio?view=vs-2022>
- II. Youtube :-
<https://www.youtube.com/watch?v=zfz49gjnjcY>